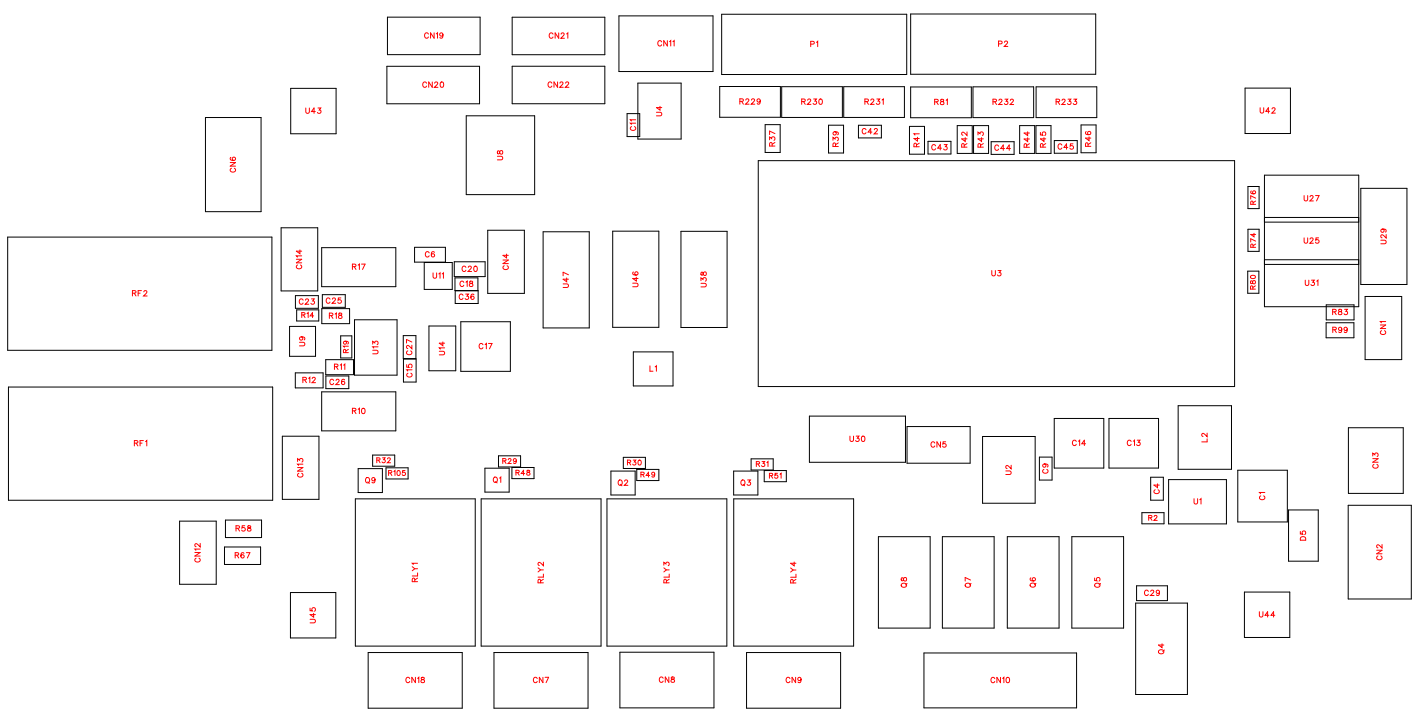
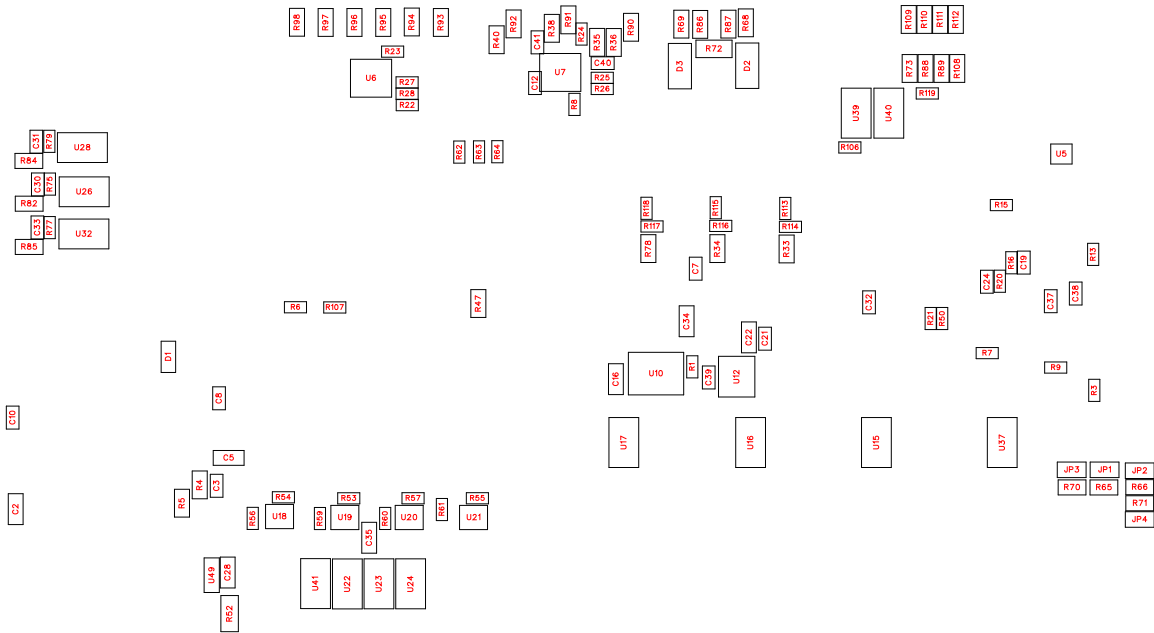






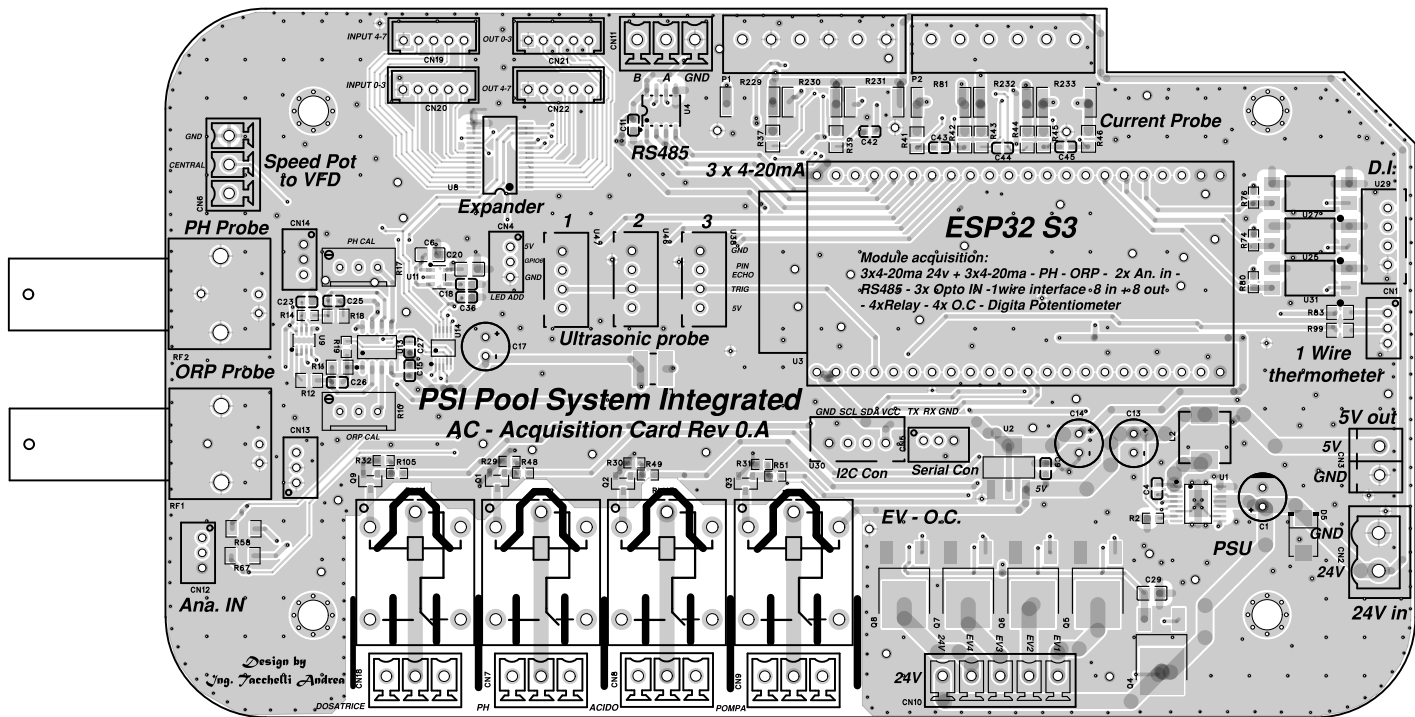
No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier
1	1	100uF	C1	CAP-TH_BD 6.3-P2.50-D0 .6-FD-H12.0	100uF	ELR1V101M 061200Y	TOPAZCON(黄宝石)	C53243582	LCSC
2	5	2.2uF	C2,C5,C6,C20,C22	C0805	2.2uF	HHV0805R7 225K101NSL J	Chinocera(华 瓷)	C22367838	LCSC
3	1	47pF	C3	C0603	47pF	0603CG470J 250NT	FH(风华)	C2939808	LCSC
4	1	470nF	C4	C0603	470nF	FCC0603B47 4K160CTM	FOJAN(富捷)	C54619744	LCSC
5	29	100nF	C7,C8,C9,C10,C11,C12,C1 5,C18,C19,C21,C23,C24,C 25,C26,C27,C30,C31,C32, C33,C36,C37,C38,C39,C40 ,C41,C42,C43,C44,C45	C0603	100nF	TCC0603X7 R104K250CT	CCTC(三环)	C7501158	LCSC
6	3	220uF	C13,C14,C17	CAP-TH_BD 6.3-P2.50-D0 .5-FD-H11.0	220uF	CD26G1AM2 21E110T	CH(创慧电子)	C51026544	LCSC
7	3	10uF	C16,C34,C35	C0805	10uF	FCC0805B10 6K160FT	FOJAN(富捷)	C54619708	LCSC
8	2	100nF	C28,C29	C0805	100nF	CC0805KRX 7R9BB104	YAGEO(国巨)	C49678	LCSC
9	5	B3B-PH-K-S(LF)(SN)	CN1,CN5,CN12,CN13,CN1 4	CONN-TH_B 3B-PH-K-S		B3B-PH-K-S(LF)(SN)	JST	C131339	LCSC
10	1	ALIMENTAZI ONE	CN2	CONN-TH WJ2EDGVC- 5.08-2P		ALIMENTAZI ONE	KANGNEX(康奈克斯电气)	C8445	LCSC
11	1	KF2EDGV-3. 81-2P	CN3	CONN-TH_2 P-P3.81_KF2 EDGV-3.81-2 P		KF2EDGV-3. 81-2P	KEFA(科发)	C441332	LCSC
12	1	LED OUT	CN4	CONN-TH_B 3B-PH-K-S		LED OUT	JST	C131339	LCSC
13	1	vfd pot	CN6	CONN-TH_Z X-DGVC-3.8 1-3P		vfd pot	LAILAN(莱联)	C55144766	LCSC
14	1	ACIDO	CN7	CONN-TH_Z X-DGVC-3.8 1-3P		ACIDO	LAILAN(莱联)	C55144766	LCSC
15	1	PH	CN8	CONN-TH_Z X-DGVC-3.8 1-3P		PH	LAILAN(莱联)	C55144766	LCSC
16	1	FILTRAGGIO	CN9	CONN-TH_Z X-DGVC-3.8 1-3P		FILTRAGGIO	LAILAN(莱联)	C55144766	LCSC
17	1	EELETTROV ALVOLE	CN10	CONN-TH_ WJ15EDGV C-3.81-5P		EELETTROV ALVOLE	KANGNEX(康奈克斯电气)	C8391	LCSC

18	1	RS485A	CN11	CONN-TH_Z X-DGVC-3.8 1-3P		RS485A	LAILAN(莱联)	C55144766	LCSC
19	1	DOSATRICE	CN18	CONN-TH_Z X-DGVC-3.8 1-3P		DOSATRICE	LAILAN(莱联)	C55144766	LCSC
20	4	PH-5A	CN19,CN20,CN21,CN22	CONN-TH_5 P-P2.00_HC TL_PH-5A_C 2908626		PH-5A	HCTL(华灿天 禄)	C2908626	LCSC
21	1	K24	D1	SOD-123FL_ L2.8-W1.8-L S3.8-RD		K24	DDF(鼎德丰)	C50229773	LCSC
22	2	SMAJ12CA	D2,D3	SMA_L4.2-W 2.7-LS5.0-BI		SMAJ12CA	GOODWOR K(固得沃克)	C908818	LCSC
23	1	SMBJ30A	D5	SMB_L4.6-W 3.6-LS5.3-R D		SMBJ30A	MDD(辰达半 导体)	C113998	LCSC
24	4	JUMPER 0805	JP1,JP2,JP3,JP4	JUMPERPA D0805					Volatile
25	1	220uH 16 OHM	L1	IND-SMD_L4 .5-W4.0_SC D0403T	220uH 16 OHM	SCD0403T-2 21K-N	Chilisin(奇力 新)		LCSC
26	1	15uH	L2	IND-SMD_L7 .7-W6.6_APS 0630	15uH	APS0630M1 50A	Coilank(驰兴 电感)	C47327208	LCSC
27	2	3x20mA	P1,P2	CONN-TH_6 P_WJ15EDG VC-3.81-6P		3x20mA	KANGNEX(康奈克斯电气)	C8415	LCSC
28	1	Q ACIDO	Q1	SOT-23-3_L2 .9-W1.3-P1.9 0-LS2.4-BR		Q ACIDO	JSMSEMI(杰 盛微)	C5296725	LCSC
29	3	Q PH	Q2,Q3,Q9	SOT-23-3_L2 .9-W1.3-P1.9 0-LS2.4-BR		Q PH	JSMSEMI(杰 盛微)	C5296725	LCSC
30	5	50N06	Q4,Q5,Q6,Q7,Q8	TO-252-2_L6 .6-W6.1-P4.5 7-LS9.9-BR- CW		50N06	HL(富海微)	C18164387	LCSC
31	41	10kΩ	R1,R3,R6,R7,R8,R9,R13,R 14,R15,R16,R19,R20,R21, R22,R23,R24,R25,R26,R27 ,R28,R29,R30,R31,R32,R4 8,R49,R50,R51,R62,R63,R 64,R74,R76,R80,R105,R10 6,R107,R113,R115,R118,R 119	R0603	10kΩ	RC0603FR-0 710KL	YAGEO(国巨)	C98220	LCSC
32	1	80.6kΩ	R2	R0603	80.6kΩ	FRC0603F80 62TS	FOJAN(富捷)	C2933260	LCSC
33	1	470kΩ	R4	R0805	470kΩ	RT0805BRE 07470KL	YAGEO(国巨)	C706280	LCSC

34	1	120k Ω	R5	R0805	120k Ω	ERA6AEB12 4V	PANASONIC (松下)	C473315	LCSC
35	2	100k Ω	R10,R17	RES-ADJ-TH 3296W-1-20 3	100k Ω	3296W-1-104	JIERR(捷而 瑞)	C48997936	LCSC
36	1	33k Ω	R11	R0805	33k Ω	0805W8F330 2T5E	UNI-ROYAL(厚声)	C17633	LCSC
37	1	15k Ω	R12	R0805	15k Ω	RT0805BRD 0715KL	YAGEO(国巨)	C728642	LCSC
38	1	4.99k Ω	R18	R0805	4.99k Ω	FRC0805F49 91TS	FOJAN(富捷)	C2933454	LCSC
39	14	4.7k Ω	R33,R34,R47,R68,R69,R73 ,R78,R88,R89,R108,R109, R110,R111,R112	R0805	4.7k Ω	0805W8F470 1T5E	UNI-ROYAL(厚声)	C17673	LCSC
40	12	10 Ω	R35,R36,R37,R38,R39,R40 ,R41,R42,R43,R44,R45,R4 6	R0805_NEW	10 Ω	0805W8F100 JT5E	UNI-ROYAL(厚声)	C17415	LCSC
41	1	10k Ω	R52	R1206	10k Ω	FRC1206F10 02TS	FOJAN(富捷)	C2907370	LCSC
42	8	100 Ω	R53,R54,R55,R56,R57,R59 ,R60,R61	R0603	100 Ω	FRC0603F10 00TS	FOJAN(富捷)	C2906981	LCSC
43	2	205 Ω	R58,R67	R1206	205 Ω	AC1206FR-0 7205RL	YAGEO(国巨)	C229437	LCSC
44	2	2k Ω	R65,R70	R0805	2k Ω	FRC0805F20 01TS	FOJAN(富捷)	C2907248	LCSC
45	2	3.3k Ω	R66,R71	R0805	3.3k Ω	FRC0805F33 01TS	FOJAN(富捷)	C2907249	LCSC
46	1	120 Ω	R72	R1206	120 Ω	FRC1206F12 00TS	FOJAN(富捷)	C2933584	LCSC
47	3	475 Ω	R75,R77,R79	R0603	475 Ω	RS-03K4750 FT	FH(风华)	C304757	LCSC
48	6	5.6 Ω	R81,R229,R230,R231,R232 ,R233	R2512	5.6 Ω	25121WF560 KT4E	UNI-ROYAL(厚声)	C620886	LCSC
49	16	0 Ω	R82,R83,R84,R85,R86,R87 ,R90,R91,R92,R93,R94,R9 5,R96,R97,R98,R99	R0805	0 Ω	0805W8F000 0T5E	UNI-ROYAL(厚声)	C17477	LCSC
50	3	4.7k Ω	R114,R116,R117	R0603	4.7k Ω	RC0603FR-0 74K7L	YAGEO(国巨)	C99782	LCSC
51	1	BNC PH	RF1	BNC-TH_DO SIN-801-003 8	BNC PH	DOSIN-801-0 038	dosinconn(德 索五金)		LCSC
52	1	3GHz	RF2	BNC-TH_DO SIN-801-003 8	3GHz	DOSIN-801-0 038	dosinconn(德 索五金)	C709673	LCSC
53	1	RELE DOSATORE	RLY1	RELAY-TH_ SRD-XXVDC -XL-C		RELE DOSATORE	松乐	C35449	LCSC
54	1	RELE PH	RLY2	RELAY-TH_ SRD-XXVDC -XL-C		RELE PH	松乐	C35449	LCSC

55	1	RELE ACIDO	RLY3	RELAY-TH SRD-XXVDC -XL-C		RELE ACIDO	松乐	C35449	LCSC
56	1	RELE FILTRAGGIO	RLY4	RELAY-TH SRD-XXVDC -XL-C		RELE FILTRAGGIO	松乐	C35449	LCSC
57	1	LM46002PW PR	U1	HTSSOP-16 L5.0-W4.4- P0.65-LS6.4- BL-EP		LM46002PW PR	TI(德州仪器)	C79086	LCSC
58	1	BL9110-330 BPFB	U2	SOT-223 L6. 5-W3.5-P2.3 0-LS7.0-BR		BL9110-330 BPFB	BL(上海贝岭)	C169246	LCSC
59	1	ESP32-S3-D EVKIT-N16R 8 COPY	U3	CONN-TH_E SP32-S3-DE VKITC-1U-N 16R8		ESP32-S3-D EVKITC-1-N 8	ESPRESSIF(乐鑫)	C20438410	
60	1	MAX3485AE ASA+T	U4	SO-8 L4.9- W3.9-P1.27- LS5.9-BL		MAX3485AE ASA+T	ADI(亚德诺)/ MAXIM(美信)	C94609	LCSC
61	1	MCP4018T-5 02E/LT	U5	SC-70-6 L2. 2-W1.3-P0.6 5-LS2.1-BR		MCP4018T-5 02E/LT	MICROCHIP(美国微芯)	C129156	LCSC
62	2	INA3221AQR GVRQ1	U6,U7	VQFN-16 L4 .0-W4.0-P0.6 5-TL-EP2.2		INA3221AQR GVRQ1	TI(德州仪器)	C1849473	LCSC
63	1	MCP23017-E /SS	U8	SSOP-28 L1 0.2-W5.3-P0. 65-LS7.8-BL		MCP23017-E /SS	MICROCHIP(美国微芯)	C506653	LCSC
64	1	OPA2338EA/ 3K	U9	SOT-23-8 L2 .9-W1.6-P0.6 5-LS2.8-BR		OPA2338EA/ 3K	TI(德州仪器)	C2877593	LCSC
65	1	LMC7660M/T R	U10	SOP-8 L4.9- W3.9-P1.27- LS6.0-BL		LMC7660M/T R	HGSEMI(华 冠)	C434489	LCSC
66	1	TMI6030-33	U11	SOT-23-5 L3 .0-W1.7-P0.9 5-LS2.8-BL		TMI6030-33	TMI(拓尔微)	C708883	LCSC
67	1	XC62KN330 2PR-G	U12	SOT-89-3 L4 .5-W2.5-P1.5 0-LS4.2-BR		XC62KN330 2PR-G	TOREX(特瑞 仕)	C	LCSC
68	1	JRC4558	U13	SOP-8 L4.9- W3.9-P1.27- LS6.0-BL		JRC4558	IDCHIP(英锐 芯)	C5795563	LCSC
69	1	ADS1115IDG SR	U14	MSOP-10 L3 .0-W3.0-P0.5 0-LS5.0-BL		ADS1115IDG SR	TI(德州仪器)	C37593	LCSC
70	13	1N4007	U15,U16,U17,U22,U23,U24 ,U26,U28,U32,U37,U39,U4 0,U41	DO-214AC L 4.3-W3.6-LS 5.3-RD		1N4007	MSKSEMI(美 森科)	C7528875	LCSC

71	4	UCC27517D BVR	U18,U19,U20,U21	SOT-23-5_L3 .0-W1.7-P0.9 5-LS2.8-BR		UCC27517D BVR	TI(德州仪器)	C99395	LCSC
72	3	PC817B-FEL -S	U25,U27,U31	SOP-4_L6.5- W4.6-P2.54- LS10.2-TL		PC817B-FEL -S	DDF(鼎德丰)	C50229497	LCSC
73	5	B4B-XH-A(L F)(SN)	U29,U30,U38,U46,U47	CONN-TH_B 4B-XH-A-LF- SN		B4B-XH-A(L F)(SN)	JST	C144395	LCSC
74	4	M4	U42,U43,U44,U45	M4 HOLE					
75	1	BZT52C16	U49	SOD-123_L2 .7-W1.6-LS3. 8-RD		BZT52C16	GOODWOR K(固得沃克)	C2903915	LCSC



Design by
Ing. Tacchelli Andrea

3.3V
DATA
GND

5V

GND

GND

24V

GND
CENTRAL
UPPER

A-IN 1
A-IN 2
GND

JP1-JP2 CLOSE= 4-20mA
JP2-JP2 OPEN= 0-10V
JP3-JP4 CLOSE= 4-20mA
JP3-JP4 CLOSE= 0-10V

NC COM NO NC COM NO NC COM NO NC COM NO

